



Master's Degree Course in **Cybersecurity**
Department of Control and Computer Engineering

An incremental approach to automatic firewall configuration

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Abstract

Nowadays, computer networks are complex systems. Modern networks are in fact often distributed systems in which security policy enforcement is a critical and continuous operational challenge. As topologies grow in scale and diversity, the manual configuration of packet-filtering firewalls becomes increasingly prone to errors. Misconfigurations can expose sensitive resources or degrade service availability. In this context, the shift toward microsegmentation, distributing enforcement across multiple firewall instances throughout the network, improves security granularity but amplifies the management effort. Automated tools that generate formally correct and optimal configurations are therefore essential.

VEREFOO (VERification REasoning on Firewall Optimization) addresses this need by automatically computing firewall allocation and configuration from a network topology and a set of Network Security Requirements (NSRs). It encodes the problem as a formal optimisation problem specifically a Maximum Satisfiability Modulo Theories (MaxSMT) instance. This approach provides two key guarantees: formal correctness, meaning that the generated configuration provably satisfies every specified requirement, and minimality, meaning that the number of allocated firewall instances and configured rules is kept as low as possible. However, VEREFOO follows a monolithic approach: every time the NSR set changes, the entire problem is re-encoded and solved from scratch, scaling poorly as the network or requirement set grows.

React-VEREFOO addresses this by taking an incremental approach. It identifies which requirements are new, removed, or unchanged, and restricts the solver to only the affected parts of the network. This reduces reconfiguration cost when few requirements change, but the benefit decreases when a large fraction of the NSR set is modified simultaneously.

This thesis pushes the incremental principle to its extreme. The proposed Iterative React-VEREFOO introduces new NSRs one at a time: at each step, the current configuration is used as the starting state and exactly one requirement is added. React-VEREFOO computes the minimal update, and the result becomes the input to the next iteration.

By construction, the reconsidered portion of the network is always minimal, keeping each MaxSMT subproblem small regardless of the total number of requirements.

The approach is evaluated on a set of benchmark topologies covering both real-world and synthetic networks of varying scale, with the aim of quantifying whether the consistent reduction in per-step problem size translates into a measurable improvement in total execution time compared to the monolithic VEREFOO baseline.